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## (Data Analytics & Reporting)

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**Course:-B.tech CS(DS+AI) 33,34**

# AIRLINES FLIGHT DATA ANALYSIS

- A Data Analytics Project using Python and Google Colab
- Name: [Shakshi Shukla/Adarsh Mishra ]
- Tools Used: Google Drive, GoogleColab,Python, Pandas,Matplotlib

# INTRODUCTION

- The airline industry generates huge amounts of data daily.
- Flight data analysis helps to improve efficiency, safety, and customer satisfaction.
- This project focuses on analyzing flight data to find useful insights.

# OBJECTIVE OF THE PROJECT

- Understand flight delays and cancellations.
- Identify the busiest airports and airlines.
- Analyze flight trends by month, day, and time.
- Visualize important patterns in the dataset.

# DATASET DOWNLOAD FROM KAGGLE

- The dataset was downloaded from Kaggle, a popular data science platform.
- The dataset includes flight details such as Airline Name, Origin and Destination, Departure and Arrival Times, Delay Information.

# UPLOAD DATASET TO GOOGLE DRIVE

- After downloading, the dataset was uploaded to Google Drive.
- This helps to access the dataset easily from anywhere.
- Google Drive acts as a storage and data source fo Google Colab.

# CONNECT GOOGLE DRIVE TO GOOGLE COLAB

- Open Google Colab.
- Mount Google Drive using the command:
- from google.colab import drive
- drive.mount('/content/drive')
- This allows Colab to access the dataset stored in Drive.

# IMPORT LIBRARIES AND LOAD DATA

- Import necessary Python libraries:
- Import pandas as pd
- Import numpy as np
- Import matplotlib.pyplot as plt
- Import seaborn as sns
- Load the dataset using Pandas:
- df = pd.read_csv('/content/drive/MyDrive/flight_data.csv')

# DATA CLEANING

- Checked for missing and duplicate values.
- Removed unnecessary columns.
- Converted date and time formats.
- Handled null or incorrect entries.

# DATA ANALYSIS

- Performed descriptive statistics using df.describe().
- Found most delayed airlines and routes.
- Grouped flights by month, day, and airline.
- Used correlation to find relationships between delay time and distance.

# DATA VISUALIZATION

- Created graphs using Matplotlib and Seaborn:
- Bar charts for flight counts per airline.
- Pie chart for flight status distribution.
- Line chart for monthly flight trends.
- Example:
- sns.barplot(x='AIRLINE', y='DEPARTURE_DELAY', data=df)
- plt.title('Average Departure Delay by Airline')
- plt.show()

# INSIGHTS & CONCLUSIONS

- Some airlines have higher average delays than others.
- Peak delays occur during certain months and holidays.
- Distance and weather also affect delay times.
- Data analysis helps airlines improve performance and customer experience.

# TOOLS AND TECHNOLOGIES USED

- Kaggle – For dataset source
- Google Drive – For storing dataset
- Google Colab – For coding and analysis
- Python Libraries – Pandas, NumPy, Matplotlib, Seaborn

# SUMMARY

- Successfully analyzed airline flight data step-by-step.
- Found useful patterns and visualized key insights.
- Demonstrated the use of cloud tools for data science workflow.

# THANK YOU

- Thank you for watching!
- Questions are welcome. ✈️